

Fixed Income Allocation Strategy

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Outline

- 1 Introduction
- 2 Methodology
- 3 Implementation: Monthly Rebalancing Strategy
- 4 Results and Performance
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Fixed Income Markets and Momentum

Core Premise: Fixed income markets exhibit time-varying risk premia that can be systematically captured through momentum-based strategies

Why momentum works in fixed income:

- Credit spreads widen or tighten gradually as economic conditions evolve, creating persistent price trends
- Interest rate regimes exhibit persistence due to central bank policy cycles and slow-moving macroeconomic fundamentals
- Rather than exploiting behavioral inefficiencies, momentum signals capture the systematic adjustment of bond prices to fundamental risk factors
- Multi-horizon BMOM framework targets these dynamics: 60-120 day lookbacks align to credit cycle frequencies, 240-day lookbacks confirm longer-term interest rate regime shifts

Asset Universe and Benchmark

Sample Period: December 1997 – November 2025 (27 years, 7,023 daily observations)

Three Fixed Income Asset Classes

- **H5A4 (High Yield):** ICE BofA US High Yield Index – non-investment-grade corporate bonds
- **C5A0 (Investment Grade):** ICE BofA US Investment Grade Corporate Index – investment-grade corporate debt
- **G5O2 (Government):** ICE BofA US Treasury Index – U.S. Treasury bonds as defensive anchor
- **Cash:** 5-10% buffer depending on market conditions

Benchmark Allocation

Asset	Weight
C5A0 (Investment Grade)	66.5%
H5A4 (High Yield)	9.5%
G5O2 (Government)	19.5%
Cash	5.0%

Regentfund's baseline static allocation that the tactical framework seeks to outperform

Implementation Constraints

Two critical constraints shape the final implementation:

1. High Yield Cap

- Maximum 12% allocation to H5A4 (high yield)
- Limits credit concentration risk in the Regentfund portfolio

2. Monthly Rebalancing

- Transaction execution limited to month-end to minimize implementation costs
- Exceptions permitted for immediate risk-off signals

Strategy Evolution

This research documents the development process through four stages:

Strategy 1: Cross-Asset Momentum Filter

- Establishes binary long/flat signals through cross-asset benchmark filtering and long-term trend confirmation

Strategy 2: Drawdown Differential Override

- Introduces state machine logic to capitalize on periods when tactical signals successfully avoid losses

Strategy 3: Proportional Linear Allocation

- Implements portfolio-level allocation across all three assets with constraints

Monthly Strategy: Operational Implementation

- Optimizes transaction timing through scheduled monthly rebalancing with risk-triggered overrides

Signal Generation: Blended Momentum (BMOM)

Foundation: Multi-horizon momentum indicator combining four distinct lookback periods

$$\text{BMOM}_i(t) = 0.15 \cdot \text{Mom}_{i,20} + 0.35 \cdot \text{Mom}_{i,60} + 0.35 \cdot \text{Mom}_{i,120} + 0.15 \cdot \text{Mom}_{i,240}$$

Asymmetric Weighting Rationale:

- **70% weight** on 60-120 day lookbacks: align with typical credit cycle frequencies and capture the gradual widening or tightening of credit spreads as economic conditions evolve
- **15% weight** on 20-day lookback: provides tactical responsiveness to recent price action
- **15% weight** on 240-day lookback: serves as regime confirmation signal, ensuring alignment with longer-term interest rate trends

This weighting structure balances responsiveness with stability, avoiding overreaction to short-term noise while remaining sensitive to persistent credit and duration dynamics

Strategy 1: Cross-Asset Momentum Filter

Key Innovation: SPY (equity market) as universal benchmark for cross-asset momentum confirmation

Signal Logic for Each Asset:

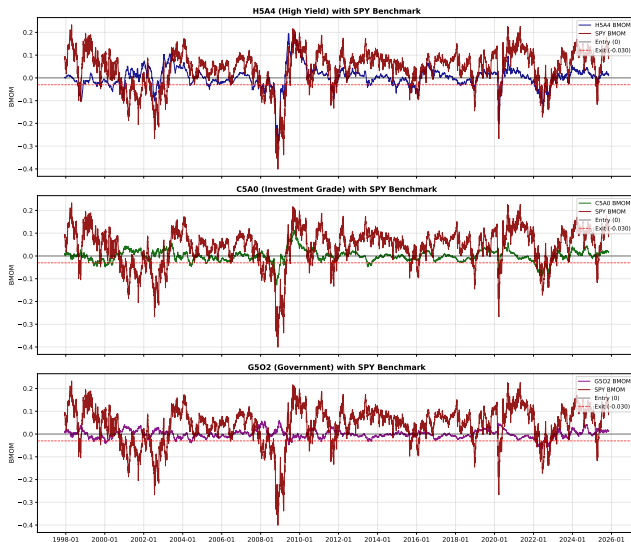
Exit Condition (Signal = 0): Exit to cash when either cross-below condition is met AND 240-day momentum is negative

Exit if: $[\text{BMOM}_{\text{asset}} \text{ and } \text{BMOM}_{\text{SPY}} \text{ both cross below } -0.03] \wedge \text{Mom}_{240} < 0$

Entry Condition (Signal = 1): $\text{BMOM}_{\text{asset}}(t) > 0$

Hold: If neither exit nor entry conditions met, maintain previous signal state

Strategy 1: BMOM Time Series with SPY Filter



Rationale:

The largest drawdowns in fixed income assets consistently overlap with periods of SPY BMOM deterioration (2008 financial crisis, 2020 COVID selloff, 2022 interest rate shock).

Dual weakness identifies systemic risk-off periods rather than idiosyncratic volatility.

The cross-asset filter exploits the equity-credit lead-lag relationship: equity markets rapidly incorporate information, while credit markets exhibit delayed price adjustment due to lower liquidity and OTC structure.

Strategy 1: Asset-Level Results

Asset	Ann. Return	Sharpe Ratio	Assessment
H5A4 (High Yield)	0.87%	0.24	Positive
C5A0 (Investment Grade)	0.54%	0.18	Positive
G5O2 (Government)	-0.42%	-0.14	Negative

Key Findings:

- Strategy 1's tactical signals successfully avoided losses in credit-sensitive assets (H5A4, C5A0)
- Opportunity identified: mechanism needed to remain invested during periods when momentum-based signals prove effective
- G5O2 failure reflects difficulty of timing duration exposure through momentum signals

Strategy 2: Drawdown Differential Override

Motivation: Capitalize on periods when tactical signals demonstrably outperform passive exposure

Mechanism: Asset-level drawdown differential override applied independently to each asset

$$DD_{\text{diff},i}(t) = DD_{S1,i}(t) - DD_{\text{BH},i}(t)$$

When $DD_{\text{diff},i} > 0.10$, Strategy 1's drawdown is more than 10 percentage points less severe than buy-and-hold, indicating the tactical signals are successfully avoiding losses

State Machine Architecture:

- ➊ **Regime Identification:** Each time $DD_{\text{diff},i}$ crosses above 0.10, a new override regime begins and receives a unique identifier
- ➋ **Override Activation:** Signal immediately forced to long (Signal = 1)
- ➌ **Regime Persistence:** Long position maintained throughout entire override regime, even if differential subsequently falls below 0.10
- ➍ **Strategic Exit:** Override regime only ends when Strategy 1 naturally signals long, ensuring exit occurs when underlying momentum logic confirms continued strength

Strategy 2: Performance Improvement

Asset	Strategy 1 Sharpe	Strategy 2 Sharpe	Improvement
H5A4 (High Yield)	0.24	0.28	+16.7%
C5A0 (Investment Grade)	0.18	0.29	+61.1%
G5O2 (Government)	-0.14	-0.14	0.0%

Critical Findings:

- The substantial improvement in C5A0 demonstrates the override's effectiveness in capturing outperformance during periods when tactical signals successfully navigate credit cycles
- No change in G5O2 indicates rare override activation for duration-focused assets
- This validates the asset class hypothesis: credit-sensitive assets exhibit exploitable momentum patterns, while Treasury timing remains challenging

Proportional Linear Allocation

Objective: Weight assets by momentum signal strength while enforcing portfolio constraints

Allocation Formula:

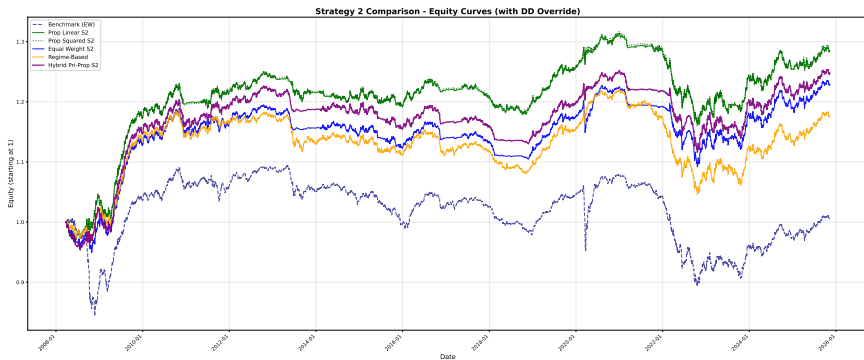
$$\text{Score}_i(t) = \max(0, \text{BMOM}_i(t)) \times \text{Signal}_i(t)$$

$$w_i^{\text{raw}} = \frac{\text{Score}_i}{\sum_j \text{Score}_j} \times 0.95 \quad (5\% \text{ cash buffer})$$

Sequential Constraint Enforcement:

- ① **Defensive Mode:** If all scores = 0, allocate 10% cash and 90% G5O2
- ② **Proportional Allocation:** Compute raw weights proportional to scores, normalized to 95%
- ③ **Minimum Position:** Active assets < 5% boosted to 5%, other assets reduced proportionally
- ④ **High Yield Cap:** If H5A4 > 12%, cap at 12% and redistribute excess to C5A0, then G5O2, then cash

Why Proportional Linear Allocation?



Comparison of allocation framework performance across the full sample period. The proportional linear allocation method (Strategy 3) demonstrated superior risk-adjusted returns compared to alternative allocation frameworks tested, while maintaining implementable portfolio constraints and preserving the tactical signal logic from Strategy 2.

Monthly Rebalancing Framework

Operational Constraint: Regentfund Fixed Income Portfolio executes trades on a monthly schedule

Dual-Trigger Approach:

1. Scheduled Monthly Rebalancing

- Portfolio weights recalculated and executed on designated monthly trading date
- Target weights determined using proportional linear allocation framework with all constraints applied

2. Risk-Triggered Rebalancing

- When any asset's Strategy 2 signal transitions to 0 (flat), immediately rebalance without waiting for next scheduled date
- Weights recalculated with flat asset receiving zero allocation, remaining active assets reweighted proportionally
- If all asset signals turn flat simultaneously, shift to defensive mode (10% cash, 90% government bonds)

Between rebalancing events, portfolio weights drift with market movements

Monthly Strategy Performance



Monthly Strategy Metrics:

Metric	Value
Annualized Return	0.77%
Annualized Vol	2.97%
Sharpe Ratio	0.26
Max Drawdown	−11.99%
Win Rate	51.05%
Rebalance Days	602
Trades/Month	1.79

Both Strategy 3 and Monthly Strategy substantially outperform the static benchmark across the full sample period, with Monthly Strategy delivering consistent returns while respecting operational trading constraints.

Strategy Evolution Performance

Asset-Level Progression (Strategy 1 → Strategy 2):

Asset	Strategy 1 Sharpe	Strategy 2 Sharpe	Improvement
H5A4 (High Yield)	0.24	0.28	+16.7%
C5A0 (Investment Grade)	0.18	0.29	+61.1%
G5O2 (Government)	−0.14	−0.14	0.0%

Portfolio-Level Comparison (Strategy 3 vs. Monthly):

Strategy	Return	Sharpe	Max DD	Trades/Mo
Strategy 3 (Daily)	1.07%	0.36	−11.71%	2.79
Monthly Strategy	0.77%	0.26	−11.99%	1.79

The Monthly Strategy represents the implementable version, sacrificing some performance for operational feasibility while maintaining core tactical allocation logic through risk-triggered overrides

Conclusion

Deliverable: Monthly Rebalancing Strategy ready for implementation in Regentfund Fixed Income Portfolio

Key Observation: Much of Strategy 3's predictive power derives from risk reduction and the drawdown differential override mechanism rather than pure directional signals.

Final Strategy Characteristics:

- 0.26 Sharpe ratio, 0.77% annualized return, -11.99% maximum drawdown
- 1.79 trades per month (scheduled monthly rebalances + risk-triggered overrides)
- Exceeds benchmark by ≥ 10 basis points annualized
- Systematic risk management: cross-asset benchmark filtering identifies broad market stress, Mom240 filter prevents exits during temporary pullbacks, dual-trigger rebalancing preserves tactical responsiveness within monthly constraints
- The framework successfully navigated major market stress events including the 2008 financial crisis, 2020 COVID-19 selloff, and 2022 interest rate shock

Appendix: Additional Technical Details

Contents:

- ① BMOM Construction Details
- ② Strategy 1 Signal Logic (Pseudocode)
- ③ Strategy 2 State Machine Algorithm
- ④ Portfolio Construction Constraints
- ⑤ Performance Metrics Definitions
- ⑥ Complete Results Table

Appendix A: BMOM Construction

Implementation Details

For each asset i , compute individual momentum terms:

$$\text{Mom}_{i,n}(t) = \frac{P_i(t) - P_i(t - n)}{P_i(t - n)}$$

Weighted sum:

$$\text{BMOM}_i(t) = \sum_{n \in \{20, 60, 120, 240\}} w_n \cdot \text{Mom}_{i,n}(t)$$

Weights: $w_{20} = 0.15$, $w_{60} = 0.35$, $w_{120} = 0.35$, $w_{240} = 0.15$

Data Handling:

- All series aligned to asset index (handles missing SPY dates)
- First valid observation at $t = 240$ (longest lookback)
- Forward-fill missing values within continuous date ranges

Appendix B: Strategy 1 Signal Logic

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1: Input:  $\text{BMOM}_{\text{asset}}, \text{BMOM}_{\text{SPY}}, \text{Mom}_{240}, \text{threshold} = -0.03$ 
2: Initialize:  $\text{Signal}(t = 0) = 0$ 
3: for  $t = 1$  to  $T$  do
4:    $c_1 \leftarrow (\text{BMOM}_{\text{SPY}}(t - 1) > -0.03) \wedge (\text{BMOM}_{\text{SPY}}(t) < -0.03) \wedge (\text{BMOM}_{\text{asset}}(t) < -0.03)$ 
5:    $c_2 \leftarrow (\text{BMOM}_{\text{asset}}(t - 1) > -0.03) \wedge (\text{BMOM}_{\text{asset}}(t) < -0.03) \wedge (\text{BMOM}_{\text{SPY}}(t) < -0.03)$ 
6:   if  $(c_1 \vee c_2) \wedge (\text{Mom}_{240}(t) < 0)$  then
7:      $\text{Signal}(t) \leftarrow 0$  // Risk-off: exit to cash
8:   else if  $\text{BMOM}_{\text{asset}}(t) > 0$  then
9:      $\text{Signal}(t) \leftarrow 1$  // Risk-on: enter long
10:  else
11:     $\text{Signal}(t) \leftarrow \text{Signal}(t - 1)$  // Hold previous position
12:  end if
13: end for
14: Output:  $\text{Signal} \in \{0, 1\}^T$ 

```

Appendix C: Strategy 2 State Machine

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1: Input: Strategy 1 signals  $s_1(t)$ , drawdown differential  $DD_{\text{diff}}(t)$ 
2: Initialize: regime_id = 0,  $s_2(t) = s_1(t)$ 
3: for  $t = 1$  to  $T$  do
4:   if  $DD_{\text{diff}}(t) > 0.10 \wedge DD_{\text{diff}}(t-1) \leq 0.10$  then
5:     regime_id  $\leftarrow$  regime_id + 1    // New override regime
6:   end if
7:   if regime_id > 0 then
8:      $s_2(t) \leftarrow 1$     // Override: force long
9:     if  $s_1(t) = 1$  then
10:      regime_id  $\leftarrow$  0    // Strategic exit: end override
11:    end if
12:   else
13:      $s_2(t) \leftarrow s_1(t)$     // Follow Strategy 1
14:   end if
15: end for
16: Output: Strategy 2 signals  $s_2(t) \in \{0, 1\}^T$ 

```

Appendix D: Portfolio Constraints

Sequential Constraint Enforcement

Step 1: Defensive Mode Check

- If $\sum_i \text{Score}_i = 0$ (all signals flat or negative momentum)
- Allocate: 10% Cash, 90% G5O2
- Exit constraint logic

Step 2: Proportional Weights

- $w_i^{\text{raw}} = \frac{\text{Score}_i}{\sum_j \text{Score}_j} \times 0.95$
- 5% to cash buffer

Step 3: Minimum Position Enforcement

- For each active asset where $w_i < 0.05$: boost to 5%
- Deficit funded by proportionally reducing other active assets

Step 4: High Yield Cap

- If $w_{\text{H5A4}} > 0.12$: cap at 12%
- Excess redistributed: first to C5A0, then G5O2, finally cash

Appendix E: Performance Metrics

Definitions

Annualized Return:

$$r_{\text{ann}} = \left(\frac{V_T}{V_0} \right)^{252/T} - 1$$

Annualized Volatility:

$$\sigma_{\text{ann}} = \sigma_{\text{daily}} \times \sqrt{252}$$

Sharpe Ratio: (Assuming zero risk-free rate)

$$\text{SR} = \frac{r_{\text{ann}}}{\sigma_{\text{ann}}}$$

Maximum Drawdown:

$$\text{DD}_{\text{max}} = \max_t \left(1 - \frac{V_t}{\max_{s \leq t} V_s} \right)$$

Win Rate:

$$\text{WR} = \frac{\#\{t : r_t > 0\}}{T}$$

Appendix F: Complete Results Table

Strategy	Return	Vol	Sharpe	Max DD	Win Rate	Trades/Mo
<i>Strategy 1 (Asset-Level):</i>						
H5A4	0.87%	3.6%	0.24	—	—	—
C5A0	0.54%	3.0%	0.18	—	—	—
G5O2	−0.42%	3.0%	−0.14	—	—	—
<i>Strategy 2 (Asset-Level):</i>						
H5A4	1.20%	4.3%	0.28	—	—	—
C5A0	0.90%	3.1%	0.29	—	—	—
G5O2	−0.42%	3.0%	−0.14	—	—	—
<i>Portfolio Strategies:</i>						
Strategy 3	1.07%	2.95%	0.36	−11.71%	50.83%	2.79
Monthly	0.77%	2.97%	0.26	−11.99%	51.05%	1.79
<i>Benchmark:</i>						
Static	—	—	—	—	—	0.00

Note: Benchmark = 66.5% C5A0, 19.5% G5O2, 9.5% H5A4, 5% Cash